

Technology Stack

Date	15 March 2026
Team ID	6aba3aa9ca93a27509644a39
Project Name	Comic Craft AI
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js with Tailwind CSS	Component-based, reusable UI for the story editor, character picker and comic preview. Responsive layout works on both desktop and mobile.

2	Backend / Server-Side	Python with FastAPI	Fast and easy to connect with AI model APIs and image-processing libraries. Handles user requests, generation jobs and comic logic.
3	Database / Data Storage	MongoDB Atlas	Stories, character profiles, panel details and user data are flexible, document-style records that change often, so a NoSQL database fits well.
4	Cloud / Hosting / Deployment	AWS (EC2 and S3) with Docker	Scales easily as users grow. S3 stores generated panel images and exported PDF/PNG files; Docker gives consistent, repeatable deployment.
5	Version Control & CI/CD	GitHub with GitHub Actions	Smooth team collaboration with branches and pull requests. Actions runs automated tests and deploys each update.

6	Third-Party APIs / Other Tools	Stable Diffusion API, Gemini API, Google Cloud Translation API	Stable Diffusion API generates the comic panel images. Gemini API turns the story script into a panel-by-panel storyboard. Translation API enables multilingual speech bubbles.